

Sreehari Jayaram

Phone (+49) 172 779 4348
Mail sriharijayaram5@gmail.com
Address Silcherstr. 76, 73614, Schorndorf

Experimental physicist with a focus on quantum technologies. Academic and commercial skills in prototyping, instrumentation and integrated system development.



confocal microscopy – cryogenics – nanofabrication – research instrumentation

Education

Ph.D. Physics – Quantum Sensing

Oct. 2021 ~ Aug. 2026

3. Physikalisches Institut, Stuttgart, Germany

Topic: Low temperature scanning probe microscopy with NV centers in diamond

Master of Science (M.Sc.) Physics – IMPRS-CMS Scholarship

2019 - 2021

University of Stuttgart, Stuttgart, Germany

Focus: Solid state physics. Thesis: Development of NV-based wide-field magnetometer for the study of correlated systems

Bachelor of Science (B.Sc.) Physics, Mathematics, Electronics

2016 - 2019

Christ (Deemed to be University), Bengaluru, India

Work experience

Microscope development

–

optics – cryogenics – instrumentation – software dev.

Widefield microscopy system - NV centers

2019 - 2021

Cryogenic widefield NV magnetometer with super-resolution (structured illumination microscopy)

Scanning probe microscopy system - NV centers

2021

Cryogenic and room temperature scanning probe NV magnetometers

Nanofabrication

–

RIE-ICP/wet etching – e-beam/UV lithography – fab. process dev.

Diamond probes

2025

Ideation, design, and fabrication of novel structures for use as atomic force microscopy probes

Product development - Squtec/TTI GmbH

–

prototyping – project management – user-centric design

Diamond Quantum Microscope for Fraunhofer IISB

2023

Development and delivery of commercial cryogenic system for spin and optical characterisation of quantum emitters in diamond and silicon carbide

Diamond Quantum Computer for Fraunhofer IAF

2024

Development of optical system and housing, and delivery of commercial 5-qubit quantum computer based on NV centers

Publications

1. **Jayaram, S.** *et al.* Probing Vortex Dynamics in 2D Superconductors with Scanning Quantum Microscope. *Phys. Rev. Lett.* **135**, 126001 (2025) – Editors' Suggestion and PRL Collection of the Year 2025
2. Wong, K.C., *et al.* Super-moiré spin textures in twisted two-dimensional antiferromagnets. *Nat. Nanotechnol.* (2026)
3. Zhou, X. *et al.* Optically Addressable Molecular Spins at 2D Surfaces. **arXiv:2601.19988** (2026)
4. Bushmakina, V. *et al.* Two-Photon Interference of Photons from Remote Tin-Vacancy Centers in Diamond. **arXiv:2412.17539** (2025)
5. Zhu, Y.-P. *et al.* Observation of plaid-like spin splitting in a noncoplanar antiferromagnet. *Nature* **626**, 523-528 (2024)
6. Prabhu, S., **Jayaram, S.**, Bubbly, S. G. & Gudennavar, S. B. A simple software for swift computation of photon and charged particle interaction parameters: PAGEX. *Applied Radiation and Isotopes* **176**, 109903 (2021)

Patent

- **Jayaram, S.** *et al.* (2026). *Method for the production of a diamond probe*. EU Patent Application No. 26 152 955.6. Filed 20.01.26

Language

- **English:** C1 (CEFR Level)
- **German:** B1 (CEFR Level)
- **Malayalam:** Native

Software

- **Programming:** skilled in Python (instrumentation, frontend, data analysis) github.com/sriharijayaram5
- **Simulation:** basic knowledge in Tidy3D (FDTD), COMSOL
- **CAD:** basic knowledge in Inventor
- **Misc.:** K-Layout (GDS), KiCad